

The Wang Method Series

Notebook #6 (Version 1.0)

Understanding the Last Two Centers of the 5×5 Rubik's Cube

A Memory-Oriented Framework

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1. Motivation

Many excellent tutorials explain how to solve the last two centers of the 5×5 Rubik's Cube. These methods are highly effective and are particularly suitable for speedcubers who are willing to memorize many algorithms in order to minimize move count.

This notebook has a different objective.

For many recreational cubers, building the first four centers is relatively straightforward, whereas the last two centers often become the most difficult stage of center solving. The difficulty is not manipulating the cube itself, but recognizing numerous patterns and recalling the appropriate algorithms after weeks or months without practice.

Rather than minimizing the total number of moves, the primary goal of this notebook is to minimize the number of concepts and algorithms that must be remembered.

This notebook therefore develops a **memory-oriented framework** based on understanding, transformation, and pattern reduction. The objective is to replace memorization with logical reasoning whenever possible.

Readers may choose between two different learning styles.

- **Move-oriented solving**, which memorizes more direct algorithms to reduce move count.
- **Memory-oriented solving**, which uses a small number of primitive operations together with simple transformations.

Neither approach is universally better. Competitive speedcubers naturally prefer shorter solutions, whereas recreational cubers and senior hobbyists may find that remembering fewer algorithms is a much more practical long-term strategy.

2. Reducing the Complexity

At first glance, the last two centers appear to be an overwhelming problem.

Each remaining center contains nine pieces. Consequently, the two unsolved centers contain eighteen positions occupied by nine green pieces and nine orange pieces.

Even when pieces of the same color are considered identical, the number of possible color arrangements is

$$\binom{18}{9} = \frac{18!}{9!9!} = 48,620$$

Clearly, no one can recognize or memorize so many configurations.

Fortunately, we do not need to.

The first reduction is to construct one complete three-piece middle line in each center.

After these two middle lines have been completed, six center pieces become fixed, leaving only twelve unresolved positions. The number of possible color arrangements immediately decreases to

$$\binom{12}{6} = \frac{12!}{6! 6!} = 924$$

Although this is a dramatic reduction, 924 configurations are still far too many for practical pattern recognition.

The second reduction is to construct another complete three-piece line, either on the left side or on the right side of each center.

At this stage, each center contains only one unfinished three-piece right line. Altogether, only six center pieces remain unresolved: three green pieces and three orange pieces.

The remaining search space therefore contains only

$$\binom{6}{3} = \frac{6!}{3! 3!} = 20$$

possible color arrangements.

This reduction from

48,620

↓

924

↓

20

forms the central idea of this notebook.

Instead of attempting to recognize tens of thousands of center configurations, we first construct stable three-piece lines that reduce the remaining search space to only twenty reduced states.

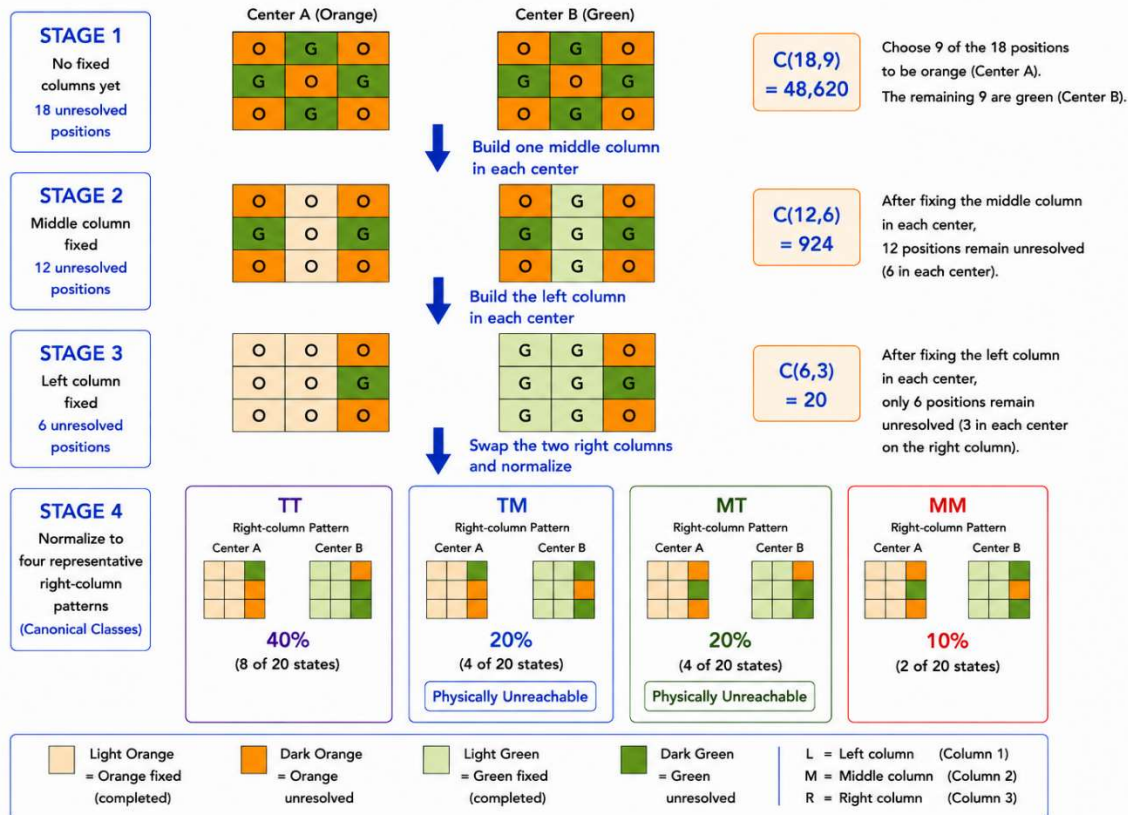


Figure 6.1: Complexity reduction from 48,620 → 924 → 20, together with the combinatorial state distribution.

3. The Twenty Reduced States

Throughout this notebook, the orange center remains on the **top face** and the green center remains on the **front face**.

After two complete three-piece lines have been constructed, only one unfinished right line remains in each center.

Rather than using binary notation, this notebook directly represents the unfinished right lines by their colors:

- **O** denotes an orange center piece.
- **G** denotes a green center piece.

For example,

- **OOO** represents a solved orange right line.
- **GGG** represents a solved green right line.

This notation is considerably more intuitive than binary notation because readers can immediately visualize the remaining center pieces without mentally translating 0s and 1s into colors.

Each unfinished right line contains three positions:

- Top (T)
- Middle (M)
- Bottom (B)

Using this representation, the twenty reduced states obtained after constructing the first two three-piece lines are summarized in Table 6.1.

Table 6.1. The Twenty Reduced Last-Two-Center States.

Case	Original state	Original errors G/O	After right-line swap	Errors after swap G/O	After optional 180° rotation for each B to T	Normalized errors G/O
1	OOO / GGG	— / —	No swap	— / —	No rotation	Solved
2	OOG / OGG	B / T	No swap	B / T	GOO / OGG	T / T
3	OOG / GOG	B / M	No swap	B / M	GOO / GOG	T / M
4	OOG / GGO	B / B	No swap	B / B	GOO / OGG	T / T
5	OGO / OGG	M / T	No swap	M / T	No rotation	M / T
6	OGO / GOG	M / M	No swap	M / M	No rotation	M / M
7	OGO / GGO	M / B	No swap	M / B	OGO / OGG	M / T
8	GOO / OGG	T / T	No swap	T / T	No rotation	T / T
9	GOO / GOG	T / M	No swap	T / M	No rotation	T / M
10	GOO / GGO	T / B	No swap	T / B	GOO / OGG	T / T
11	OGG / OOG	MB / TM	OOG / OGG	B / T	GOO / OGG	T / T
12	OGG / OGO	MB / TB	OGO / OGG	M / T	No rotation	M / T
13	OGG / GOO	MB / MB	GOO / OGG	T / T	No rotation	T / T
14	GOG / OOG	TB / TM	OOG / GOG	B / M	GOO / GOG	T / M
15	GOG / OGO	TB / TB	OGO / GOG	M / M	No rotation	M / M
16	GOG / GOO	TB / MB	GOO / GOG	T / M	No rotation	T / M
17	GGO / OOG	TM / TM	OOG / GGO	B / B	GOO / OGG	T / T
18	GGO / OGO	TM / TB	OGO / GGO	M / B	OGO / OGG	M / T
19	GGO / GOO	TM / MB	GOO / GGO	T / B	GGO / OGG	T / T
20	GGG / OOO	TMB / TMB	OOO / GGG	— / —	No rotation	Solved

Instead of analyzing 48,620 possible center configurations, we now need to understand only twenty reduced states.

These twenty states are not equally important.

After applying right-line swaps and simple cube rotations for normalization, they naturally fall into four canonical unsolved classes together with the solved state.

The occurrence frequencies are summarized in Table 6.2.

Table 6.2. Occurrence frequencies of 20 states.

Class	Number of Cases	Fraction of 20 States
Solved	2	10%
TT	8	40%
TM	4	20%
MT	4	20%
MM	2	10%

Several interesting observations immediately follow.

First, the **TT** family represents 8 of the 20 combinatorial reduced states, or 40% of the abstract state space.

Second, the **MM** family represents 2 of the 20 states, or 10% of the abstract state space.

Finally, although the combinatorial model produces TM and MT configurations, these configurations are not physically reachable on a real 5×5 cube. The reduced color model does not preserve the fixed inventory of physical center pieces. For example, some TM and MT configurations would require a total of three or five corner pieces of one color across the last two centers, whereas a real 5×5 cube has exactly four. Thus, a configuration may be combinatorially valid in the reduced color model without being physically realizable on the cube.

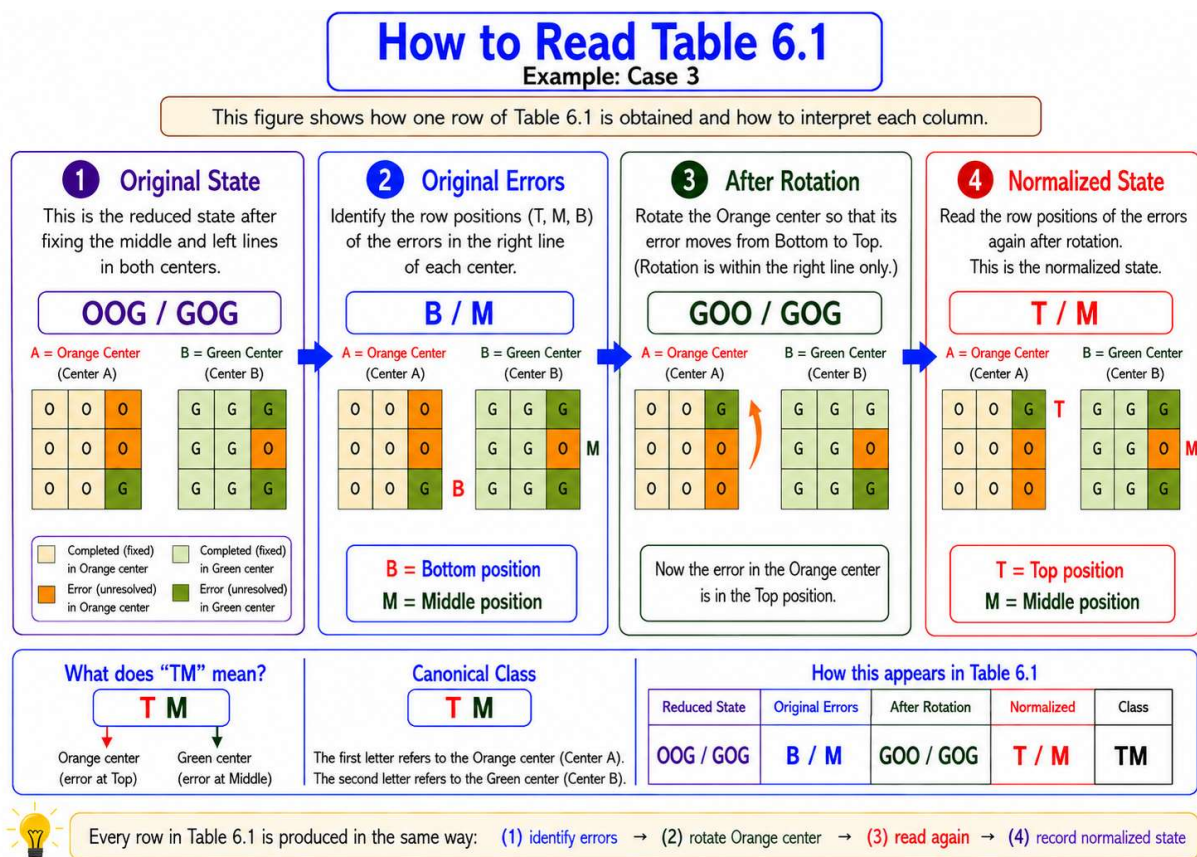


Figure 6.2: How to read Table 6.1 and illustrate the swap and rotation operations.

4. From Twenty Reduced States to Four Canonical Classes

The twenty reduced states are not independent.

The first simplification is obtained by exchanging the two unfinished right lines. This simple operation transforms Cases 11–20 into Cases 1–10. Consequently, only the first ten cases need to be analyzed.

The second simplification is achieved by a simple setup rotation.

Whenever an error appears at the bottom position (B), the corresponding center layer may be rotated so that the bottom error is moved to the top position (T). This normalization does not change the relative relationship between the two unfinished right lines; it merely provides a common reference orientation.

Consequently,

- TT includes TT, TB, BT, and BB.
- TM includes TM and BM.
- MT includes MT and MB.
- MM remains unchanged.

After applying these two normalization procedures, the original twenty reduced states are classified into only four canonical unsolved classes together with the solved state.

The normalization process is illustrated in Figure 6.3, which shows how right-line swaps and simple setup rotations reduce twenty reduced states into only four canonical classes.

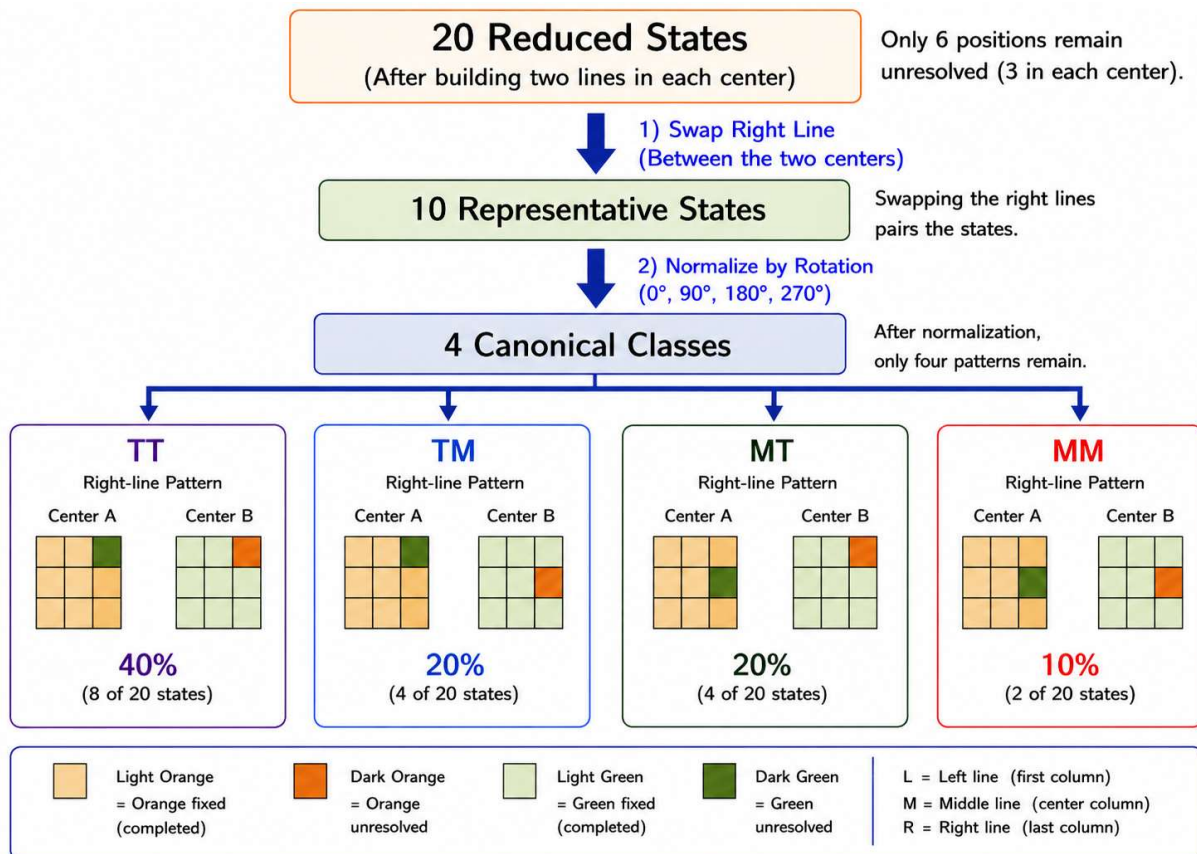


Figure 6.3: Reduction from 20 reduced states to Solved, TT, TM, MT, and MM.

This classification provides a much simpler framework for understanding the last two centers than treating all twenty states independently.

5. Two Alternative Solutions for the TT Family

Among the four canonical classes, the TT family is particularly important because it represents 8 of the 20 combinatorial reduced states.

Two fundamentally different solution strategies are available.

Memory-Oriented Transformation

The first strategy minimizes memorization.

Instead of directly applying a seven-move primitive, the TB case is first transformed by exchanging the two unfinished right lines.

Depending on the orientation, this transformation produces one of two simple local configurations.

Each configuration can then be solved using the familiar four-move local sequence

$U' R w U R w'$

or its equivalent front-face version

$F' R w' F R w$.

Although this approach requires approximately fourteen moves in total, every step is local, intuitive, and easy to reconstruct after long periods without practice.

The complete procedure is illustrated in Figure 6.4.

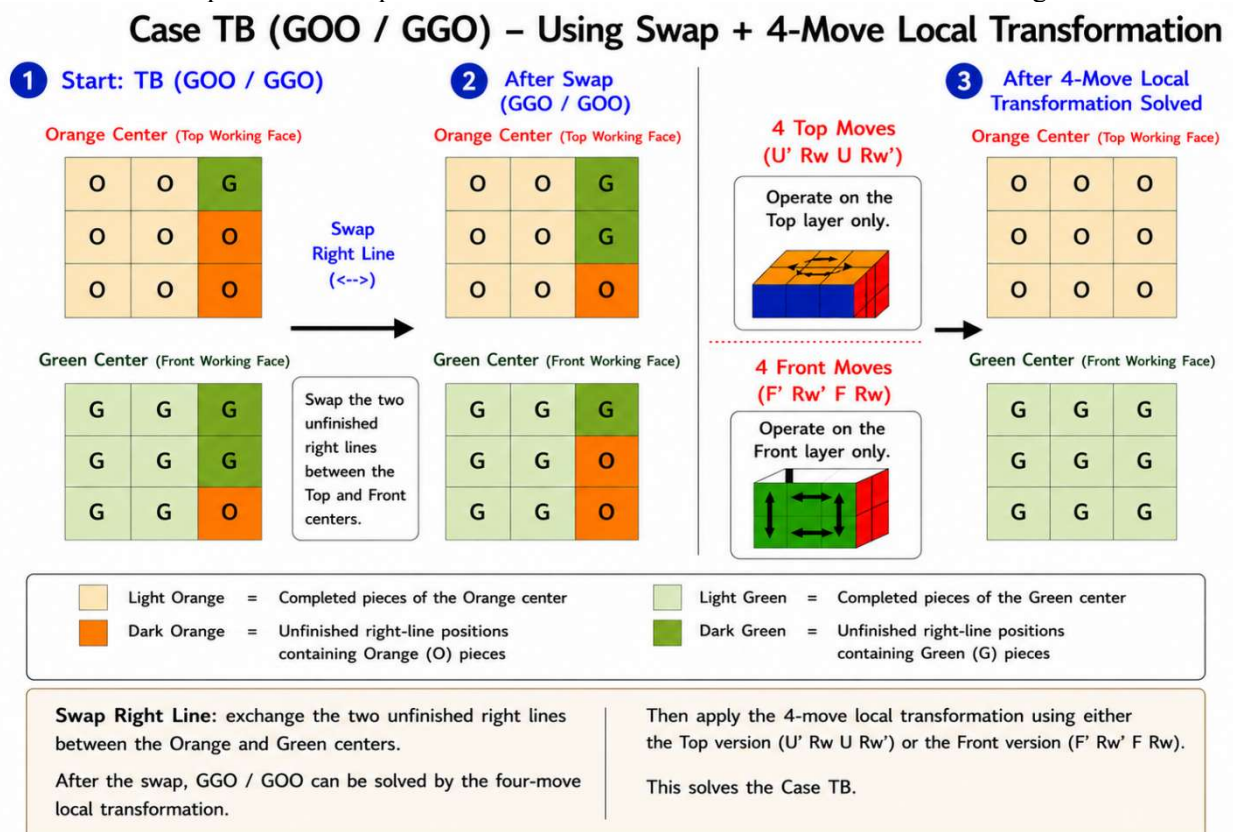


Figure 6.4: Using swap and 4-move local transformation to solve TB (GOO/GGO).

Direct TT Primitive

The second strategy minimizes move count. **The direct TT and MM move sequences are commonly referred to as commutators in the cubing community.**

With the orange center on the top face and the green center on the front face, one representative TT primitive is

Rw' F' Lw F Rw F' Lw'

requiring approximately seven moves.

Unlike the previous method, this primitive temporarily borrows workspace from a neighboring completed center before restoring it at the end of the sequence.

Rather than viewing this simply as a memorized algorithm, it is helpful to understand its three functional stages:

1. Borrow temporary workspace.
2. Transfer the required center pieces.
3. Restore the borrowed workspace.

Understanding these functions makes the primitive easier to reconstruct, even if the exact move sequence is temporarily forgotten.

The two approaches therefore represent different design philosophies.

The direct primitive minimizes move count. The transformation method minimizes memory burden. Readers may choose whichever approach best matches their own learning style.

As another practical option, cubers may transform a TT state to BT and then apply the right Sune algorithm. This may be especially convenient for cubers who are already familiar with the right Sune for solving OLL corners during 4×4 CFOP.

More generally, cubers do not need to memorize a separate direct algorithm for every case. An unfamiliar case may instead be transformed into a familiar case whose algorithm is already known. The preferred transformation is therefore not necessarily the one with the fewest setup moves, but one that provides a favorable tradeoff between move count and memory burden. In some cases, a transformation may add only one or a few moves while eliminating the need to memorize an additional algorithm.

For example, the TB/TB case can be transformed to MM in four moves and then solved with the seven-move MM algorithm, for a total of 11 moves—the same move count as its direct solution. Thus, cubers who already know the MM algorithm need not memorize a separate TB/TB algorithm.

6. Why TM and MT Do Not Occur Physically

The combinatorial reduction generates four TM states and four MT states among the twenty reduced color arrangements. However, these eight states are not physically reachable on a real 5×5 cube.

The reason is that the reduced model tracks only the colors appearing in the two remaining centers; it does not preserve the fixed inventory of the physical center pieces. For example, some of the abstract TM and MT arrangements would require a total of three or five corner pieces of one color across the last two centers, rather than the four physically available on the cube. Such arrangements can therefore exist in the combinatorial model but cannot occur as legal physical states.

This explains why TM and MT appear naturally in the twenty-state combinatorial classification even though they are not encountered when solving a real cube. It also illustrates an important distinction between a **combinatorially valid color arrangement** and a **physically reachable cube state**.

7. Two Learning Paths

Throughout this investigation, two complementary approaches for solving the last two centers have emerged.

The first is a **move-oriented approach**, which minimizes the total number of moves by memorizing a larger collection of direct algorithms.

The second is the **memory-oriented approach** developed in this notebook, which minimizes the number of algorithms by using simple transformations together with a small number of primitive operations.

These two approaches represent different learning philosophies rather than competing methods.

For speedcubers who solve the cube every day, memorizing additional algorithms may be entirely reasonable if it reduces move count.

For recreational cubers, however, the situation is often quite different.

After already learning OLL, PLL, and parity algorithms, many hobbyists may find it difficult to memorize another large collection of L2C algorithms and retain them after weeks or months without practice.

The memory-oriented framework therefore seeks a different balance.

Instead of asking,

"Which algorithm should I memorize?"

it asks,

"Can this configuration first be transformed into one that I already understand?"

This philosophy trades a modest increase in move count for a substantial reduction in memory burden. Readers may therefore choose whichever approach best matches their own goals and learning preferences.

For convenient use during practice, the complete set of four L2C reference cards is provided in the Appendix.

8. Lessons Learned

This investigation began with a simple question:

Why are the last two centers so difficult?

At first glance, the answer appeared to be the large number of possible patterns and the many algorithms required to solve them.

The analysis presented in this notebook suggests a different explanation. The true difficulty lies not in the enormous number of theoretical configurations, but in the limited temporary workspace available after most center pieces have already been completed.

By progressively constructing stable three-piece lines, the search space is reduced from

$$\binom{18}{9} = \frac{18!}{9! 9!} = 48,620$$

to

$$\binom{12}{6} = \frac{12!}{6! 6!} = 924$$

and finally to

$$\binom{6}{3} = \frac{6!}{3! 3!} = 20$$

The remaining twenty reduced states can then be understood systematically through swap and rotation, leading to only a small number of canonical solution families. More importantly, this study illustrates a broader principle. A seemingly complicated problem often becomes much simpler after identifying the right representation.

Rather than relying primarily on memorization, one may instead search for transformations that convert unfamiliar situations into familiar ones. This viewpoint forms the central philosophy of the memory-oriented framework presented in this notebook.

9. Concluding Remarks

This notebook does not attempt to present the fastest solution for the last two centers of the 5×5 Rubik's Cube. Instead, it proposes a different way of thinking about the problem.

Its principal contribution is not a new algorithm, but a systematic framework that reduces complexity through mathematical analysis, simple transformations, and a small number of primitive operations.

Readers who prefer the shortest solutions may choose to memorize a larger collection of direct algorithms. Readers who prefer a lighter memory burden may instead adopt the memory-oriented approach developed here.

Both approaches are valid. They simply optimize different objectives.

Acknowledgments

The author gratefully acknowledges Marty Kelsey (Martyt7777) for his valuable insights and helpful discussions on center-solving algorithms, which helped inspire the systematic study of the 5×5 last-two-center cases presented in this notebook. The author also thanks members of the SpeedSolving.com community who generously shared references, links, technical comments, and their cubing experience during the development of this work.

Appendix. L2C Reference Cards

The following four reference cards summarize the last two centers (L2C) cases and algorithms discussed in this notebook. They are included as a practical companion to the combinatorial classification developed in the main text and are intended for quick reference during solving and practice.

Reference Cards – Representative Cases

Direct Solutions for the Last Two Centers (L2C)

CASE 1

SOLVED
(OOO / GGG)

Orange Center
(Top Working Face)

O	O	O
O	O	O
O	O	O

Green Center
(Front Working Face)

G	G	G
G	G	G
G	G	G

Direct Solution

Already Solved
(No moves needed)

CASE 8

TT
(GOO / OGG)

Orange Center
(Top Working Face)

O	O	G
O	O	O
O	O	O

Green Center
(Front Working Face)

G	G	O
G	G	G
G	G	G

Direct Solution

Rw' F' Lw F Rw F' Lw'
(7 moves)

CASE 6

MM
(OGO / GOG)

Orange Center
(Top Working Face)

O	O	O
O	O	G
O	O	O

Green Center
(Front Working Face)

G	G	G
G	G	O
G	G	G

Direct Solution

Rw U' M' U Rw' U' M
(7 moves)

= Correctly placed Orange center pieces

= Misplaced (unfinished) positions containing Orange (O) pieces

= Correctly placed Green center pieces

= Misplaced (unfinished) positions containing Green (G) pieces

Note: T = Top; M = Middle; B = Bottom

Figure 6.5: Possible algorithms to solve Cases 1, 8, and 6.

Reference Cards – Representative Cases

Direct Solutions for the Last Two Centers (L2C)

CASE 2

BT
(OOG / OGG)

Orange Center
(Top Working Face)

O	O	O
O	O	O
O	O	G

Green Center
(Front Working Face)

G	G	O
G	G	G
G	G	G

Direct Solution

(Rw U Rw') U (Rw U2' Rw')

CASE 4

BB
(OOG / GGO)

Orange Center
(Top Working Face)

O	O	O
O	O	O
O	O	G

Green Center
(Front Working Face)

G	G	G
G	G	G
G	G	O

Direct Solution

Rw' F Lw F' Rw F Lw'

CASE 10

TB
(GOO / GGO)

Orange Center
(Top Working Face)

O	O	G
O	O	O
O	O	O

Green Center
(Front Working Face)

G	G	G
G	G	G
G	G	O

Direct Solution

(Rw U' Rw') U' (Rw U2' Rw')

= Correctly placed Orange center pieces

= Misplaced (unfinished) positions containing Orange (O) pieces

= Correctly placed Green center pieces

= Misplaced (unfinished) positions containing Green (G) pieces

Note: T = Top; M = Middle; B = Bottom

Figure 6.6: Possible algorithms to solve Cases 2, 4, and 10.

Reference Cards – Representative Cases

Direct Solutions for the Last Two Centers (L2C)

CASE 11

MB / TM → TT
(OGG / OOG)

Orange Center
(Top Working Face)

O	O	O
O	O	G
O	O	G

Green Center
(Front Working Face)

G	G	O
G	G	O
G	G	G

Direct Solution

U (Rw U' Rw')

CASE 13

MB / MB → TT
(OGG / GOO)

Orange Center
(Top Working Face)

O	O	O
O	O	G
O	O	G

Green Center
(Front Working Face)

G	G	G
G	G	O
G	G	O

Direct Solution

U (Rw U Rw') U' (Rw U' Rw')

CASE 17

TM / TM → TT
(GGG / OOG)

Orange Center
(Top Working Face)

O	O	G
O	O	G
O	O	O

Green Center
(Front Working Face)

G	G	O
G	G	O
G	G	G

Direct Solution

U' (Rw U' Rw') U (Rw U Rw')

= Correctly placed Orange center pieces

= Misplaced (unfinished) positions containing Orange (O) pieces

= Correctly placed Green center pieces

= Misplaced (unfinished) positions containing Green (G) pieces

Note: T = Top; M = Middle; B = Bottom

Figure 6.7: Possible algorithms to solve Cases 11, 13, and 17.

Reference Cards – Representative Cases

Direct Solutions for the Last Two Centers (L2C)

CASE 19

TM / MB → TT
(GGG / GOO)

Orange Center
(Top Working Face)

O	O	G
O	O	G
O	O	O

Green Center
(Front Working Face)

G	G	G
G	G	O
G	G	O

Direct Solution

U' (Rw U Rw')

CASE 15

TB / TB → MM
(GOG / OGO)

Orange Center
(Top Working Face)

O	O	G
O	O	O
O	O	G

Green Center
(Front Working Face)

G	G	O
G	G	G
G	G	O

Direct Solution

(Rw U' Rw' U) (Rw U' Rw' U) (Rw U' Rw')

CASE 20

TMB / TMB → SOLVED
(GGG / OOO)

Orange Center
(Top Working Face)

O	O	G
O	O	G
O	O	G

Green Center
(Front Working Face)

G	G	O
G	G	O
G	G	O

Direct Solution

U2 (Rw U2 Rw')

= Correctly placed Orange center pieces

= Misplaced (unfinished) positions containing Orange (O) pieces

= Correctly placed Green center pieces

= Misplaced (unfinished) positions containing Green (G) pieces

Note: T = Top; M = Middle; B = Bottom

Figure 6.8: Possible algorithms to solve Cases 19, 15, and 20.